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# Vector Balancing via Directional Total Variation

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**Abstract**

Our main result is a $3\sqrt{2\pi}$ bound for the Komlós signing problem: every finite family of real vectors of Euclidean norm at most one admits a signed sum of $\ell_\infty$-norm less than this constant, independently of the dimension and the family size. For any $\kappa \geq 0$, if a bounded open convex set supports a probability density with directional total variation at most $\kappa$ in every unit direction, then its open-set Banaszczyk transform supports another such density with the same $\kappa$, provided the translation vector $v$ satisfies $\kappa\|v\|_2 \leq 1/3$. As a consequence, every finite set system in which each element belongs to at most $t$ sets, where $t \geq 1$ is an integer, admits a two-coloring whose imbalance in each set is less than $3\sqrt{2\pi t}$. This gives the square-root dependence predicted by the Beck–Fiala conjecture. The proof was discovered by the Odin Automatic AI Research Agent.

## 1 Introduction

Discrepancy theory asks how evenly a collection of objects can be two-colored while balancing many tests at once. For a set system on $\{1, \ldots, n\}$ with incidence matrix $A \in \{0,1\}^{m\times n}$, a coloring $\varepsilon \in \{-1,1\}^n$ gives the signed set imbalances as the entries of $A\varepsilon$. Allowing real entries gives weighted tests and the definition

$$\operatorname{disc}(A) = \min_{\varepsilon\in\{-1,1\}^n} \|A\varepsilon\|_\infty, \qquad A \in \mathbb{R}^{m\times n}.$$

Up to a factor of two, discrepancy measures the error in simultaneously rounding the all-1/2 vector. If $\mathbf{1}_n \in \mathbb{R}^n$ is the all-one vector and $x = (\mathbf{1}_n + \varepsilon)/2$, then $A(x - \frac{1}{2}\mathbf{1}_n) = \frac{1}{2}A\varepsilon$. See Bansal, Dadush, and Garg [BDG19, Section 1] for this connection with optimization.

The Komlós conjecture asks whether $\operatorname{disc}(A)$ is bounded by a universal constant whenever every column of $A$ has Euclidean norm at most one. This hypothesis bounds each vector's total squared contribution to the tests, with no restriction on the number of tests or vectors. Our main result is the following bound.

**Theorem 1.1.** *For every pair of positive integers $m, n$ and every $v_1, \ldots, v_n \in \mathbb{R}^m$ with $\|v_j\|_2 \leq 1$, there are $\varepsilon_1, \ldots, \varepsilon_n \in \{-1,1\}$ such that*

$$\left\| \sum_{j=1}^{n} \varepsilon_j v_j \right\|_\infty < 3\sqrt{2\pi}. \tag{1.1}$$

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The Euclidean normalization includes bounded-degree set systems. Its consequence is the square-root dependence in the Beck–Fiala conjecture [BF81]; the formulation recalled by Bansal, Dadush, and Garg [BDG19, Section 1] uses the maximum element degree.

**Corollary 1.2** (Beck–Fiala bound)**.** *Let $m, n, t$ be positive integers. If $A \in \{0,1\}^{m\times n}$ has at most $t$ nonzero entries in each column, then*

$$\operatorname{disc}(A) < 3\sqrt{2\pi t}.$$

*Proof.* Every column of $A/\sqrt{t}$ has Euclidean norm at most one, so Theorem 1.1 gives

$$\operatorname{disc}(A) = \sqrt{t}\,\operatorname{disc}(A/\sqrt{t}) < 3\sqrt{2\pi t}. \qquad \square$$

The argument is existential. No polynomial-time implementation of the recursive domain construction, its membership queries, and the resulting sign recovery is established here. We do not claim that the constant is optimal.

## 1.1 Development of the Problem

Beck and Fiala [BF81] obtained the classical linear bound $2t - 1$; see the later account [BDG19, Section 1]. Spencer's *six standard deviations* theorem [Spe85] gives $6\sqrt{n}$ for $n$ sets on $n$ elements, as stated by Alon and Spencer [AS16, Theorem 13.2.1]. The partial-coloring method fixes a positive fraction of the variables at each stage. Lovett and Meka [LM15, Sections 1–2] give a constructive version yielding the $O(\sqrt{n})$ order in this regime. Under a column norm or degree bound, however, progress in the number of colored variables need not imply comparable progress for each row; see Bansal, Dadush, and Garg [BDG19, Section 1.2]. Summing the classical partial-coloring estimates gives $O(\log n)$ for Komlós, as reviewed by Reis and Rothvoss [RR23, Section 1].

Banaszczyk's vector-balancing theorem [Ban98] introduced a Gaussian-measure criterion. In the formulation recalled by Dadush, Garg, Lovett, and Nikolov [DGLN19, Theorem 1.1], vectors of Euclidean norm at most $1/5$ can be signed into any convex body of standard Gaussian measure at least $1/2$. A cube of fixed half-side length has Gaussian measure tending to zero with $m$. The direct cube bound is $O(\sqrt{\log m})$; the refined Komlós consequence is $O(\sqrt{\log n})$, uniformly in $m$, as explained by Dadush et al. [DGLN19, Section 1, p. 3]. Bansal, Dadush, and Garg [BDG19, Theorem 2] obtained an efficient randomized algorithm attaining this latter bound for matrices whose columns have Euclidean norm at most one. The Gram–Schmidt walk of Bansal, Dadush, Garg, and Lovett samples full colorings with a constant sub-Gaussian bound on the discrepancy vector relative to the initial fractional coloring. Together with the convex-body reduction, this gives an algorithmic form of the general Gaussian-measure theorem, up to an absolute constant [BDGL19, Theorems 1.2 and 1.4].

Bansal and Jiang [BJ26a, Theorem 1.2, full version] establish the Komlós bound $\widetilde{O}((\log n)^{1/4})$, where the tilde suppresses powers of $\log\log n$. Their subsequent exposition [BJ26b, Theorem 1.1] states the explicit bound

$$O\big((\log n)^{1/4}(\log\log n)^{7/4}\big).$$

The exposition does not optimize the secondary factor. Its proof controls interactions among rows through affine spectral independence [BJ26b, Section 1.1].

For Beck–Fiala, Altschuler and Tikhomirov [AT26, Corollary 1.2] obtain $O_\eta(\sqrt{t})$ when $t \geq C_\eta(\log n)(\log\log n)^{2+\eta}$, for fixed $\eta > 0$ and sufficiently large $n$. Their online theorem gives probabilistic prefix guarantees for input sequences fixed in advance [AT26, Theorem 1.1]. Our Corollary 1.2 applies to every positive integer $t$, without a lower restriction depending on $n$.

Nikolov [Nik13, Theorem 1.1] proved the constant-one bound for *vector discrepancy*: each scalar sign is replaced by a unit vector, and each row sum is measured in Euclidean norm. Reis and Rothvoss [RR23, Theorems 1–2] study balancing from $\ell_p$ into $\ell_q$, with their partial-to-full result excluding $(p, q) = (2, \infty)$. Guillen and Kobzar [GK26, Theorem 1.1 and Section 1.1] connect a repeated online balancing game to curvature flow in a long-time limit, a different regime from a single finite signing.

In the opposite direction, Kunisky [Kun23, Theorem 1.1] constructs matrices with column Euclidean norms at most one whose discrepancy approaches $1 + \sqrt{2}$.

### 1.2 The Variation Approach

Our geometric framework is an open-set version of Banaszczyk's convex-body transform [Ban98]. Its retained fibers expand while the transformed set stays in $(K - v) \cup (K + v)$, permitting sign recovery. Szusterman [Szu25, Section 1] recalls the original transform and this recovery argument. The Gaussian-threshold variant in that paper interacts with Ehrhard symmetrization [Szu25, Section 2]. We use the strict Euclidean fiber-length threshold in (2.4).

For a smooth compactly supported density $\rho$, $V_u(\rho) = \int |\partial_u \rho|$ is its first-order $L^1$-translation rate in direction $u$. Small variation means that a short translation leaves most mass overlapping its previous position. The extension to functions of bounded variation (BV) includes boundary jumps. For $\kappa \geq 0$, we require one density with $V_u(\rho) \leq \kappa \|u\|_2$ for every $u$, so that the same bound applies to vectors in unrelated directions.

Smirnov and Vershynin [SV26, Theorem 1.1 and Section 1.4] also optimize a density, using translation Fisher information to bound expected discards from a fixed input sequence with independent random signs. Their online algorithm accepts or discards each incoming term; the accepted partial sums remain in twice the symmetric convex body supporting the density. Their proof [SV26, Section 2] already relates translation overlap to directional $L^1$-variation before bounding it by Fisher information. Here the main step preserves a common directional-variation bound as the target domain changes, allowing every input vector to receive a sign. We control the final sum; a corresponding bound for every prefix is not obtained here. Their product cosine-squared density [SV26, Section 1.6] gives our initial bound by a direct $L^1$ estimate.

The main analytic step transfers averaged horizontal variation bounds in a symmetric convex lift to a prescribed section. For a fixed weighted sum of directional variations, the uniform density on a set of finite perimeter and positive volume has energy equal to its anisotropic perimeter divided by its volume. Minimizing the energy over admissible probability densities gives an anisotropic Cheeger value. Eigenvalue convexity, following Wang and Xia [WX11, Theorem 1.2], and a $p \downarrow 1$ limit make that cost convex along the sections. Appendix A supplies the eigenvalue argument for the smooth, strongly convex norms used in the approximation. Jensen's inequality then bounds this value at the mass-weighted mean absolute height of the lift. Convex separation and compactness reconstruct a single density controlling all directions on that section. An exact rearrangement identity ensures that height one is available when $\kappa \|v\|_2 \leq 1/3$. Thus the transform retains the same numerical bound, although the density can change at every step.

**The role of AI in this proof.** Odin Automatic AI Research Agent was used to construct the proof.

**Paper organization.** Section 2 reduces full signing to finite-step stability and estimates the initial cube density. Section 3 establishes the prescribed-section principle through Cheeger convexity and a common-density argument. Section 4 combines that principle with a geometric lift and fiber rearrangement to prove stability.

## 2 From Variation Stability to Full Signings

All function spaces are over the real numbers. Integrable functions and their identities are understood up to equality almost everywhere. Unspecified integrals and all derivatives are taken on the ambient Euclidean space. For $f \in L^1(\mathbb{R}^d)$ and $u \in \mathbb{R}^d$, define

$$V_u(f) = \sup_{\substack{\phi \in C_c^1(\mathbb{R}^d) \\ \|\phi\|_\infty \le 1}} \int_{\mathbb{R}^d} f \, \partial_u \phi \in [0, \infty]. \tag{2.1}$$

When finite, this is the total variation of the distributional derivative $D_u f$, namely $|D_u f|(\mathbb{R}^d)$. Placing an absolute value around the integral gives the same supremum, since the test class is closed under $\phi \mapsto -\phi$. For $f \in W^{1,1}(\mathbb{R}^d)$, it equals $\int |\partial_u f|$. We use $BV(\mathbb{R}^d)$ for the integrable functions whose full distributional gradient is a finite vector measure, and write $|Df|(\mathbb{R}^d)$ for its total variation. For probability densities $f, g$, the total variation *distance* between their laws is a different quantity:

$$d_{\mathrm{TV}}(f\,dx, g\,dx) = \tfrac{1}{2}\|f - g\|_1. \tag{2.2}$$

For a nonempty bounded open convex set $K \subset \mathbb{R}^d$, let $\mathcal{P}(K)$ be the nonnegative functions $\rho \in BV(\mathbb{R}^d)$ with

$$\int \rho = 1, \qquad \rho = 0 \quad \text{almost everywhere on } \mathbb{R}^d \setminus K.$$

Throughout, "supported in $K$" has this almost-everywhere meaning. The topological support may meet $\partial K$; compact support inside $K$ will be required only for specific smooth approximations. For $\kappa \ge 0$, the invariant is the existence of $\rho \in \mathcal{P}(K)$ satisfying

$$V_u(\rho) \le \kappa \|u\|_2 \qquad \text{for every } u \in \mathbb{R}^d. \tag{2.3}$$

The global derivative is important even for elementary densities. For $C > 0$, the density $\rho = (2C)^{-1}\mathbf{1}_{(-C,C)}$ is constant inside its support, but its two endpoint jumps give $V_1(\rho) = 1/C$. For $|h| \le 2C$, its $L^1$ translation change is exactly $|h|/C$. Computing variation only inside the open interval would miss it.

For a vector $v \in \mathbb{R}^d$, define

$$\mathcal{T}_v K = ((K - v) \cap (K + v)) + \{tv : -2 < t < 2\}. \tag{2.4}$$

This is an open-set version of Banaszczyk's convex-body transform [Ban98], whose closed-set definition and use in sign recovery are recalled by Szusterman [Szu25, Section 1, pp. 1–2]. When $v \neq 0$, fibers parallel to $v$ of length at most $2\|v\|_2$ disappear; longer fibers grow by $\|v\|_2$ at each end. The closed-set convention retains fibers at the threshold, whereas (2.4) uses a strict inequality. For example, $\mathcal{T}_1(-1, 1) = \varnothing$, while the closed interval $[-1, 1]$ is transformed into $[-2, 2]$. Thus taking the interior of the closed-set transform need not give (2.4). The variation estimate below must control the loss of short fibers.

**Lemma 2.1** (Translate containment)**.** *For a bounded open convex set $K$, the set $\mathcal{T}_v K$, when nonempty, is bounded, open, and convex, and*

$$\mathcal{T}_v K \subseteq (K - v) \cup (K + v). \tag{2.5}$$

*If $K = -K$, then $\mathcal{T}_v K = -\mathcal{T}_v K$.*

*Proof.* The geometric properties follow from (2.4). Write $y = z + tv$, with $z - v, z + v \in K$ and $|t| < 2$. If $t \ge 0$, convexity gives $y - v = z + (t-1)v \in K$. If $t \le 0$, it gives $y + v = z + (t+1)v \in K$. This proves (2.5). □

**Proposition 2.2** (Finite-step stability)**.** *Let $K \subset \mathbb{R}^d$ be nonempty, bounded, open, and convex. Let $\kappa \geq 0$, $v \in \mathbb{R}^d$, and suppose $\rho \in \mathcal{P}(K)$ satisfies* (2.3)*. If*

$$\kappa \|v\|_2 \leq \frac{1}{3}, \tag{2.6}$$

*then $\mathcal{T}_v K$ is nonempty and admits $\widetilde{\rho} \in \mathcal{P}(\mathcal{T}_v K)$ with $V_u(\widetilde{\rho}) \leq \kappa \|u\|_2$ for every $u \in \mathbb{R}^d$.*

The proof in Section 4 only needs $V_v(\rho) \leq 1/3$ as its step condition; (2.6) ensures this under (2.3). The stated form gives the following criterion for iterated signing.

**Corollary 2.3** (Symmetric targets)**.** *Let $K \subset \mathbb{R}^d$ be nonempty, bounded, open, and convex, with $K = -K$. Suppose that, for some $\kappa \geq 0$, there is $\rho \in \mathcal{P}(K)$ satisfying* (2.3)*. For every integer $n \geq 1$ and family $v_1, \dots, v_n \in \mathbb{R}^d$ with $\kappa \max_j \|v_j\|_2 \leq 1/3$, there are signs $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$ such that*

$$\sum_{j=1}^{n} \varepsilon_j v_j \in K.$$

*Proof.* Set $K_0 = K$ and $K_j = \mathcal{T}_{v_j} K_{j-1}$ for $1 \leq j \leq n$. Proposition 2.2 applies at every stage with the same $\kappa$. Each $K_j$ is nonempty, bounded, open, and convex. Lemma 2.1 preserves symmetry about the origin. If $z \in K_n$, then $-z \in K_n$, so convexity gives $0 \in K_n$. This is the only use of symmetry in the signing reduction; the density chosen at each stage need not be symmetric.

Starting from $z_n = 0$, choose successively $\varepsilon_j \in \{-1, 1\}$, for $j = n, \dots, 1$, so that

$$z_{j-1} = z_j + \varepsilon_j v_j \in K_{j-1}.$$

Such a choice exists by (2.5). Consequently

$$\sum_{j=1}^{n} \varepsilon_j v_j = z_0 \in K. \qquad \square$$

The initial bound comes from the product cosine-squared density used by Smirnov and Vershynin [SV26, Section 1.6, proof of Corollary 1.3]. It is the square of the $L^2$-normalized first Dirichlet eigenfunction of the cube. We estimate its directional $L^1$-variation.

**Lemma 2.4** (A cube density)**.** *For $C > 0$, the probability density*

$$\rho_C(x) = C^{-d} \prod_{i=1}^{d} \cos^2\Big(\frac{\pi x_i}{2C}\Big)\, \mathbf{1}_{(-C,C)^d}(x) \tag{2.7}$$

*belongs to $\mathcal{P}((-C, C)^d)$ and satisfies*

$$V_u(\rho_C) \leq \frac{\sqrt{2\pi}}{C} \|u\|_2 \qquad (u \in \mathbb{R}^d). \tag{2.8}$$

*Proof.* Each normalized one-dimensional factor has integral one and vanishes at its endpoints. The zero extension is in $W^{1,1}(\mathbb{R}^d)$. If $X$ has density $\rho_C$, the variables $T_i = \tan(\pi X_i/(2C))$ are independent with density $2/(\pi(1+t^2)^2)$. Differentiation gives

$$V_u(\rho_C) = \frac{\pi}{C}\,\mathbb{E}\left|\sum_i u_i T_i\right|.$$

The independent $T_i$ admit the joint representation in distribution $T_i = G_i/\sqrt{S_i}$, where the $G_i$ are standard Gaussian variables and the $S_i$ have the $\chi^2_3$ distribution, all independent. Indeed,

$$\int_0^\infty \sqrt{\frac{s}{2\pi}}e^{-st^2/2}\frac{s^{1/2}e^{-s/2}}{\sqrt{2\pi}}\,\mathrm{d}s = \frac{2}{\pi(1+t^2)^2},$$

and the same gamma integral gives $\mathbb{E}S_i^{-1} = 1$. Conditioning on the $S_i$ and applying Jensen yields

$$\begin{aligned}\mathbb{E}\left|\sum_i u_i T_i\right| &= \sqrt{\frac{2}{\pi}}\,\mathbb{E}\sqrt{\sum_i \frac{u_i^2}{S_i}} \\ &\le \sqrt{\frac{2}{\pi}}\sqrt{\sum_i u_i^2 \mathbb{E}S_i^{-1}} = \sqrt{\frac{2}{\pi}}\,\|u\|_2.\end{aligned}$$

This proves (2.8). The $T_i$ are centered, with $\mathbb{E}T_i^2 = \mathbb{E}S_i^{-1} = 1$, so Cauchy–Schwarz alone would give $V_u(\rho_C) \le (\pi/C)\|u\|_2$. Conditioning on the $S_i$ improves the constant to $\sqrt{2\pi}/C$. □

*Proof of Theorem 1.1.* Set $C = 3\sqrt{2\pi}$, $K = (-C, C)^m$, and $\kappa = 1/3$. Lemma 2.4 supplies a density satisfying (2.3). Since $\|v_j\|_2 \le 1$, Corollary 2.3 gives a signing whose sum lies in $K$. Membership in this open cube gives the strict inequality in (1.1). □

*Remark* 2.5 (Sharpness for the product density). The coefficient $\sqrt{2\pi}$ in (2.8) is optimal uniformly over dimension for this density family. To see this, fix $C > 0$, write $\rho_C^{(d)}$ for (2.7) in $\mathbb{R}^d$, and take $u_d = d^{-1/2}(1, \dots, 1)$. The conditional Gaussian calculation gives

$$CV_{u_d}(\rho_C^{(d)}) = \sqrt{2\pi}\,\mathbb{E}\sqrt{\frac{1}{d}\sum_{i=1}^d S_i^{-1}},$$

where the $S_i$ are independent with law $\chi^2_3$. Since $\mathbb{E}S_i^{-1} = 1$, the strong law makes the square root tend almost surely to one. Fatou bounds the lower limit of these expectations from below by one, while Jensen bounds each expectation from above by one. Thus $CV_{u_d}(\rho_C^{(d)}) \to \sqrt{2\pi}$. This concerns the fixed product family; it does not establish optimality among all cube-supported densities or for the Komlós constant.

## 3 A Cheeger Principle for Convex Sections

The lift will control horizontal variations averaged over its sections. We seek the same bounds on one prescribed section. We first treat finite weighted sums of directional variations, then use convex separation to recover a common density.

We record the elementary variation facts used in this passage.

**Lemma 3.1** (Translation and compactness)**.** *For every $f \in L^1(\mathbb{R}^d)$, $u \in \mathbb{R}^d$, and $h \neq 0$,*

$$\|f(\cdot + hu) - f\|_1 \leq |h| V_u(f), \qquad V_u(f) = \lim_{h \to 0} \frac{\|f(\cdot + hu) - f\|_1}{|h|}, \tag{3.1}$$

*where the equality allows the value $+\infty$. The functional $V_u$ is convex and lower semicontinuous in $L^1$. Probability densities supported in a fixed bounded set and with uniformly bounded coordinate variations have an $L^1$-convergent subsequence.*

*Proof.* For finite $V_u(f)$, mollify $f$. The directional derivative of the mollification has $L^1$ norm at most $V_u(f)$. Integrating along segments and passing to the $L^1$ limit gives the first inequality. Conversely, for each test function in (2.1), integration against difference quotients and passage to the limit give

$$\left| \int f \, \partial_u \phi \right| \leq \|\phi\|_\infty \liminf_{h \to 0} \frac{\|f(\cdot + hu) - f\|_1}{|h|}.$$

Taking the supremum proves the equality, including the infinite case. Convexity and lower semicontinuity follow directly from the same test-function definition.

Coordinate bounds give a uniform bound on $|Df|(\mathbb{R}^d)$, since $|Df|(\mathbb{R}^d) \leq \sum_i V_{e_i}(f)$. Successive translations in the coordinate directions give

$$\|f(\cdot + h) - f\|_1 \leq \sum_{i=1}^{d} |h_i| V_{e_i}(f), \qquad h \in \mathbb{R}^d,$$

so the translation modulus is uniform over the family. Approximating each density by its averages on cubes of a fixed small mesh therefore gives uniformly small $L^1$ errors. These averages lie in a bounded subset of a finite-dimensional space, because the supports are bounded and the masses are one. A diagonal selection over successively finer meshes gives an $L^1$-Cauchy subsequence, proving the compactness assertion. □

For a continuous, even, convex, positively one-homogeneous function $H : \mathbb{R}^d \to [0, \infty)$, $\rho \in BV(\mathbb{R}^d)$, and a nonempty bounded open convex set $K \subset \mathbb{R}^d$, define

$$E_H(\rho) = \int_{\mathbb{R}^d} H\left( \frac{dD\rho}{d|D\rho|} \right) \mathrm{d}|D\rho|, \qquad h_H(K) = \inf_{\rho \in \mathcal{P}(K)} E_H(\rho). \tag{3.2}$$

Here $dD\rho/d|D\rho|$ is the Radon–Nikodym density of the vector measure $D\rho$ with respect to its total variation. Writing it as $\sigma_\rho$, we have

$$D_u \rho = (u \cdot \sigma_\rho)|D\rho|, \qquad V_u(\rho) = \int |u \cdot \sigma_\rho| \, \mathrm{d}|D\rho|.$$

The definition immediately gives domain monotonicity: $K \subset L$ implies $h_H(L) \leq h_H(K)$. For a bounded set $E$ with smooth boundary and positive volume, the uniform density has energy

$$E_H\left( \frac{\mathbf{1}_E}{|E|} \right) = \frac{1}{|E|} \int_{\partial E} H(\nu_E) \, \mathrm{d}\mathcal{H}^{d-1},$$

where $\nu_E$ is the unit normal and $\mathcal{H}^{d-1}$ is surface measure. Thus $H$ assigns a cost to each boundary orientation, and the normalization compares that cost with the volume carrying the mass. More generally, write $P_H(E) = \int_{\partial^* E} H(\nu_E) \, d\mathcal{H}^{d-1}$ for a set of finite perimeter, where $\partial^* E$ is its

reduced boundary. The anisotropic BV coarea formula, as used by Kawohl and Novaga [KN08, proof of Theorem 4.1], and layer cake give, for $\rho \in \mathcal{P}(K)$,

$$E_H(\rho) = \int_0^\infty P_H(\{\rho > s\})\,\mathrm{d}s, \qquad 1 = \int_0^\infty |\{\rho > s\}|\,\mathrm{d}s.$$

Almost every superlevel set at a positive level has finite perimeter. The energy is at least the infimum of the perimeter-to-volume ratios of those with positive volume. Uniform densities give the reverse inequality, so

$$h_H(K) = \inf_{\substack{|E\setminus K|=0,\ |E|>0 \\ E \text{ of finite perimeter}}} \frac{P_H(E)}{|E|}. \tag{3.3}$$

For a norm $H$, this is the anisotropic Cheeger constant of Kawohl and Novaga [KN08, equations (5), (10), and proof of Theorem 4.1]. In their notation, our gradient and normal integrand $H$ corresponds to the dual norm $\phi^*$, not to the norm $\phi$ defining distance. Perimeters are computed in $\mathbb{R}^d$, including any reduced boundary lying on $\partial K$.

We use integrands that are finite convex combinations of absolute linear forms:

$$H(\xi) = \sum_{\ell=1}^{N} \alpha_\ell |u_\ell \cdot \xi|, \qquad \|u_\ell\|_2 = 1, \quad \alpha_\ell \ge 0, \quad \sum_\ell \alpha_\ell = 1. \tag{3.4}$$

The corresponding energy is $E_H(\rho) = \sum_\ell \alpha_\ell V_{u_\ell}(\rho)$. Such an $H$ can vanish in nonzero directions and need not be a norm. The connection with first Dirichlet eigenvalues of anisotropic $p$-Laplacians is treated by Kawohl and Novaga [KN08, Theorem 4.1]. We give the limit argument for the global BV class in (3.2), then pass to possibly degenerate combinations (3.4).

**Lemma 3.2** (Minkowski convexity)**.** *For $H$ as in* (3.4)*, nonempty bounded open convex sets $K_0, K_1 \subset \mathbb{R}^d$, and $0 \le t \le 1$,*

$$h_H((1-t)K_0 + tK_1) \le (1-t)h_H(K_0) + t h_H(K_1). \tag{3.5}$$

*Proof.* First let $H$ be a norm, $C^\infty$ away from the origin, with $D^2(H^2)$ positive definite there. Thus $H$ is even, positive away from zero, and homogeneous of degree one. For $1 < p < \infty$, put

$$\lambda_{p,H}(K) = \inf_{0 \ne f \in W_0^{1,p}(K)} \frac{\int_K H(\nabla f)^p}{\int_K |f|^p}. \tag{3.6}$$

Norm equivalence and the Dirichlet Poincaré inequality make this quantity positive and finite for every nonempty bounded open $K$. With $K_t = (1-t)K_0 + tK_1$, we need

$$\lambda_{p,H}(K_t) \le (1-t)\lambda_{p,H}(K_0) + t\lambda_{p,H}(K_1). \tag{3.7}$$

This follows from Wang and Xia's inverse-power inequality [WX11, Theorem 1.2], with their integrand $F = H$ and the same Rayleigh normalization. For the stated strongly convex class, Proposition A.1 gives a proof using the eigenfunction results of Mosconi, Riey, and Squassina [MRS24, Theorem 1.1 and Proposition 4.5]. It handles critical points by a weak cutoff argument and removes smooth boundary assumptions by inner exhaustion. No differentiability of $H^2$ at zero is required.

We claim that

$$\lim_{p \downarrow 1} \lambda_{p,H}(K) = h_H(K). \tag{3.8}$$

First, smooth compactly supported probability densities suffice for the infimum in (3.2). Choose $x_0 \in K$ and $\delta_0 > 0$ with $B(x_0, \delta_0) \subset K$. For $0 < r < 1$, convexity gives

$$K_r := x_0 + r(\overline{K} - x_0) \Subset K, \qquad \operatorname{dist}(K_r, K^c) \geq (1-r)\delta_0.$$

For a feasible density $\rho$, contraction gives a probability density supported in $K_r$, with energy $r^{-1}E_H(\rho)$, namely

$$\rho_r(x) = r^{-d}\rho\left(x_0 + \frac{x - x_0}{r}\right).$$

Convolve its global zero extension with a nonnegative smooth kernel of mass one and radius less than $(1-r)\delta_0$. The result belongs to $C_c^\infty(K)$, has mass one, and has energy at most $E_H(\rho_r)$, by convexity and one-homogeneity of $H$. Letting $r \uparrow 1$ proves the assertion. This approximation includes the derivative measure on the support boundary.

For $0 \neq f \in W_0^{1,p}(K)$, its zero extension has $|f|^p \in W^{1,1}(\mathbb{R}^d)$. Testing with $\rho = |f|^p / \int |f|^p$, the chain rule and Hölder give

$$h_H(K) \leq p \left( \frac{\int H(\nabla f)^p}{\int |f|^p} \right)^{1/p}.$$

Consequently,

$$\lambda_{p,H}(K) \geq \big(h_H(K)/p\big)^p, \qquad \liminf_{p \downarrow 1} \lambda_{p,H}(K) \geq h_H(K).$$

For the upper limit, fix $\delta > 0$ and then choose a nonnegative $f_\delta \in C_c^\infty(K)$ with mass one and $E_H(f_\delta) \leq h_H(K) + \delta$. This same function is a Rayleigh test for every $p > 1$, and dominated convergence gives

$$\begin{aligned} \limsup_{p \downarrow 1} \lambda_{p,H}(K) &\leq \lim_{p \downarrow 1} \frac{\int_K H(\nabla f_\delta)^p}{\int_K f_\delta^p} \\ &= E_H(f_\delta) \leq h_H(K) + \delta. \end{aligned}$$

Let $\delta \downarrow 0$. This proves (3.8); applying it to $K_0, K_1, K_t$ in the eigenvalue inequality gives (3.5).

For the mixture (3.4) and $\epsilon > 0$, set

$$H_\epsilon(\xi) = \sum_\ell \alpha_\ell \sqrt{(u_\ell \cdot \xi)^2 + \epsilon^2 \|\xi\|_2^2}.$$

Each ellipsoidal norm in this sum is smooth away from zero and has nonnegative Hessian with kernel exactly the radial direction. Since the nonnegative weights sum to one, $D^2 H_\epsilon(\xi)$ is positive semidefinite with kernel $\operatorname{span}\{\xi\}$ for every $\xi \neq 0$. In

$$D^2(H_\epsilon^2) = 2\nabla H_\epsilon \otimes \nabla H_\epsilon + 2H_\epsilon D^2 H_\epsilon,$$

the second term is positive on every nonradial vector. On radial vectors the first term is positive, by $\nabla H_\epsilon(\xi) \cdot \xi = H_\epsilon(\xi) > 0$. Thus $D^2(H_\epsilon^2)$ is positive definite away from zero. These are precisely the smooth, strongly convex integrands covered by Proposition A.1 and the preceding limit argument. Moreover,

$$H \leq H_\epsilon \leq H + \epsilon \|\cdot\|_2.$$

For $\delta > 0$, fix $\rho_\delta \in \mathcal{P}(K)$ with $E_H(\rho_\delta) \leq h_H(K) + \delta$. Then

$$h_H(K) \leq h_{H_\epsilon}(K) \leq h_H(K) + \delta + \epsilon |D\rho_\delta|(\mathbb{R}^d).$$

First let $\epsilon \downarrow 0$ with this density fixed, and then let $\delta \downarrow 0$. It follows that $h_{H_\epsilon}(K) \to h_H(K)$. Passing to this limit at $K_0, K_1, K_t$ proves (3.5), even if the directions do not span $\mathbb{R}^d$. No uniform BV bound for minimizers of the degenerating energies is required. □

The next lemma recovers one density from bounds on all finite weighted infima. Attainment of the individual weighted infima is not required: separation followed by compactness supplies exact simultaneous directional bounds.

**Lemma 3.3** (A common density)**.** *Let $K \subset \mathbb{R}^d$ be nonempty, bounded, open, and convex. Let $\kappa \geq 0$. If $h_H(K) \leq \kappa$ for every mixture* (3.4)*, then there is $\rho \in \mathcal{P}(K)$ satisfying* (2.3)*.*

*Proof.* Fix finitely many unit directions $u_1, \ldots, u_N$ including the coordinate basis. Including this basis will turn bounds on the listed directions into the BV bound

$$|D\rho|(\mathbb{R}^d) \leq \sum_{i=1}^{d} V_{e_i}(\rho) \leq d \max_{1 \leq \ell \leq N} V_{u_\ell}(\rho).$$

Its dimension dependence is used only to obtain compactness in the fixed space $\mathbb{R}^d$, not in the eventual directional bound. The set

$$S = \{z \in \mathbb{R}^N : \exists \rho \in \mathcal{P}(K),\ z_\ell \geq V_{u_\ell}(\rho)\ (1 \leq \ell \leq N)\}$$

is nonempty, convex, and upward closed in every coordinate. If $\inf_{\rho \in \mathcal{P}(K)} \max_\ell V_{u_\ell}(\rho) > \kappa$, choose $a$ strictly between these values. The convex sets $S$ and $Q = (-\infty, a)^N$ are disjoint, and $Q$ is open. The separation theorem for a convex set and a disjoint open convex set provides a nonzero linear functional $\beta$ with

$$\inf_{z \in S} \beta \cdot z \geq \sup_{q \in Q} \beta \cdot q.$$

Since $S + \tau e_\ell \subset S$ for all $\tau \geq 0$, each $\beta_\ell$ must be nonnegative. Divide by $\sum_\ell \beta_\ell > 0$ to obtain weights $\alpha_\ell$. The supremum over $Q$ is now $a$, so

$$\inf_{\rho \in \mathcal{P}(K)} \sum_\ell \alpha_\ell V_{u_\ell}(\rho) \geq a > \kappa,$$

contrary to the hypothesis. Thus the finite-direction infimum is at most $\kappa$.

A minimizing sequence for the finite maximum has bounded coordinate variations. Lemma 3.1 gives a subsequence converging in $L^1$ to $\rho$. Nonnegativity and mass one pass to the limit, and the support condition follows directly from

$$\int_{K^c} |\rho| \leq \|\rho - \rho_j\|_1 \longrightarrow 0.$$

The coordinate bounds also give $\rho \in BV(\mathbb{R}^d)$. Lower semicontinuity of the finitely many variations shows that the minimum is attained, with every listed variation at most $\kappa$.

Now take increasing finite lists containing the coordinate basis and a countable dense set of unit directions. The corresponding minimizing densities satisfy $|D\rho|(\mathbb{R}^d) \leq d\kappa$. Another application of compactness gives a single limit density in $\mathcal{P}(K)$. Each fixed direction from the countable list occurs in all sufficiently late lists, so lower semicontinuity preserves its bound along this one subsequence. Finally,

$$V_u(\rho) \leq V_w(\rho) + \|u - w\|_2 \, |D\rho|(\mathbb{R}^d)$$

extends the bound to all unit $u$, and homogeneity gives (2.3). □

We now pass from averaged horizontal variation bounds to a prescribed section.

**Lemma 3.4** (Prescribed sections)**.** *Let $B \subset \mathbb{R}^d \times \mathbb{R}$ be nonempty, bounded, open, and convex, and invariant under $(y,s) \mapsto (y,-s)$. Put $D_t = \{y : (y,t) \in B\}$, $t \geq 0$. Let $\kappa \geq 0$. Suppose $R \in L^1(\mathbb{R}^{d+1})$ is a probability density, vanishes almost everywhere outside $B$, and satisfies*

$$V_{(u,0)}(R) \leq \kappa \qquad (\|u\|_2 = 1).$$

*If $a \geq 0$ and $\int |s| R \geq a$, then $D_a$ is nonempty and admits $\rho \in \mathcal{P}(D_a)$ satisfying* (2.3)*.*

*Proof.* The projection of $B$ on the last coordinate is an interval $(-b,b)$, with $0 < b < \infty$. Thus precisely the sections with $0 \leq t < b$ are nonempty. Reflection and convexity give $D_t \subseteq D_s$ for $0 \leq s \leq t < b$, and

$$(1-\theta)D_s + \theta D_t \subseteq D_{(1-\theta)s+\theta t} \qquad (0 \leq s,t < b,\ 0 \leq \theta \leq 1).$$

For a fixed mixture $H$, domain monotonicity gives that $g(t) := h_H(D_t)$ is nondecreasing. The same monotonicity and Lemma 3.2 give

$$\begin{aligned} h_H(D_{(1-\theta)s+\theta t}) &\leq h_H((1-\theta)D_s + \theta D_t) \\ &\leq (1-\theta)h_H(D_s) + \theta h_H(D_t). \end{aligned}$$

Hence $g$ is also convex on $[0,b)$. It is finite there, since each nonempty section admits a smooth compactly supported probability density.

For almost every $s$, let $R_s = R(\cdot, s)$ and $q(s) = \int_{\mathbb{R}^d} R(y,s)\,\mathrm{d}y$. For each fixed direction $u$, the difference-quotient formula (3.1), Fubini, and Fatou give

$$\begin{aligned} \int_{\mathbb{R}} V_u(R_s)\,\mathrm{d}s &= \int_{\mathbb{R}} \lim_{h\to 0} \frac{\|R_s(\cdot + hu) - R_s\|_1}{|h|}\,\mathrm{d}s \\ &\leq \liminf_{h\to 0} \frac{\|R(\cdot + (hu,0)) - R\|_1}{|h|} = V_{(u,0)}(R). \end{aligned} \tag{3.9}$$

Intersecting the full-measure sets for the finitely many coordinate directions gives $R_s \in BV(\mathbb{R}^d)$ for almost every $s$. Fubini, the support condition, and the reflection symmetry of $B$ then give $R_s/q(s) \in \mathcal{P}(D_{|s|})$ for almost every $s$ with $q(s) > 0$. If $q(s) = 0$, nonnegativity gives $R_s = 0$ almost everywhere and its energy contributes zero.

The measure $q(s)\,\mathrm{d}s$ is a probability measure concentrated on $(-b,b)$. Its mean absolute height $M = \int |s| q(s)\,\mathrm{d}s$ satisfies $M \geq a$, and

$$b - M = \int_{-b}^{b} (b - |s|) q(s)\,\mathrm{d}s > 0.$$

Thus $D_M$ and $D_a$ are nonempty, including when $M = a$. Normalization of a slice divides its energy by $q(s)$; its contribution is therefore weighted by that slice's mass. For the fixed mixture $H$, this yields

$$\int_{-b}^{b} q(s) h_H(D_{|s|})\,\mathrm{d}s \leq \sum_{\ell} \alpha_\ell \int_{\mathbb{R}} V_{u_\ell}(R_s)\,\mathrm{d}s \leq \kappa.$$

This estimate makes $g(|s|)$ integrable with respect to $q(s)\,\mathrm{d}s$. Applying Jensen under the distribution of $|s|$ induced by $q(s)\,\mathrm{d}s$, and then using its monotonicity, gives

$$\kappa \geq \int_{-b}^{b} q(s) h_H(D_{|s|})\,\mathrm{d}s \geq h_H(D_M) \geq h_H(D_a).$$

This holds for every mixture $H$. Lemma 3.3 now supplies the required density on $D_a$. □

*Remark* 3.5. Only the region $B$ must be reflection symmetric. Neither $R$ nor its marginal $q$ is assumed symmetric, and no vertical derivative of $R$ is used. The density obtained on $D_a$ is a new common density, not necessarily one of the normalized input slices.

# 4 Lifting and Finite-Step Stability

We now prove Proposition 2.2. The cutoff three makes the height-one section equal to the sign transform: an open interval in $(-3, 3)$ has length greater than two exactly when it contains $[c-1, c+1]$ for some $|c| < 2$.

**Lemma 4.1** (The geometric lift)**.** *For nonempty bounded open convex $K \subset \mathbb{R}^d$ and $v \in \mathbb{R}^d$, define*

$$B = \{(y, s) : |s| < 3,\ y + sv \in K\}.$$

*Let $\ell(y)$ be the length of its vertical fiber, zero for an empty fiber. The Steiner symmetral*

$$B^\star = \{(y, s) : |s| < \ell(y)/2\}$$

*is nonempty, bounded, open, convex, and symmetric in $s$. Its height-one section is*

$$\{y : \ell(y) > 2\} = \mathcal{T}_v K. \tag{4.1}$$

*Proof.* The lift $B$ is nonempty, bounded, open, and convex. Its nonempty vertical fibers $J_y$ are intervals, and

$$(1-\theta)J_{y_0} + \theta J_{y_1} \subseteq J_{(1-\theta)y_0+\theta y_1}.$$

Their lengths form a positive concave function on the open convex projection of $B$, hence are continuous there. This proves the stated properties of $B^\star$.

If $J_y = (\alpha, \beta)$ has length greater than two, choose $c \in (\alpha+1, \beta-1) \subseteq (-2, 2)$. Here $c$ is the center of a length-two segment in the vertical fiber, and $z = y + cv$ is its corresponding midpoint in $K$. Thus $z \pm v \in K$, so $y \in \mathcal{T}_v K$. Conversely, if $y = z + tv$, $|t| < 2$, and $z \pm v \in K$, then $-t-1, -t+1 \in J_y$. Openness gives $\ell(y) > 2$. This also covers $v = 0$. □

The next lemma relates the geometric symmetral $B^\star$ to fiberwise density rearrangement by showing how support is preserved.

**Lemma 4.2** (Fiber rearrangement)**.** *For nonnegative $R \in L^1(\mathbb{R}^d \times \mathbb{R})$, put*

$$\mu(y, t) = \int_{\mathbb{R}} \mathbf{1}_{\{R(y,s)>t\}}\, \mathrm{d}s, \qquad R^\star(y, s) = \int_0^\infty \mathbf{1}_{\{2|s|<\mu(y,t)\}}\, \mathrm{d}t, \tag{4.2}$$

*where $t > 0$, and set $R^\star = 0$ on the exceptional nonintegrable fibers. The formula is understood almost everywhere, with value zero chosen wherever it is infinite. Then $R^\star$ is jointly measurable, nonnegative, and has the same mass as $R$. If $R = 0$ almost everywhere outside the lift $B$ in Lemma 4.1, then $R^\star = 0$ almost everywhere outside $B^\star$. For every $u \in \mathbb{R}^d$ and $h \in \mathbb{R}$,*

$$\|R^\star(\cdot + (hu, 0)) - R^\star\|_1 \le \|R(\cdot + (hu, 0)) - R\|_1. \tag{4.3}$$

*Consequently $V_{(u,0)}(R^\star) \le V_{(u,0)}(R)$.*

*Proof.* Tonelli makes $\mu$ and the function defined by the integral formula measurable, and gives

$$\int_{\mathbb{R}^{d+1}} R^\star = \int_{\mathbb{R}^d} \int_0^\infty \mu(y,t)\,\mathrm{d}t\,\mathrm{d}y = \int_{\mathbb{R}^{d+1}} R.$$

The integral formula is therefore finite almost everywhere, so the chosen finite-valued representative has the same mass. For almost every $y$, the function $\mu(y,\cdot)$ is decreasing and right-continuous. Up to null sets, the superlevel sets of $R^\star(y,\cdot)$ at positive levels are centered intervals of the same lengths as those of $R(y,\cdot)$. Under the support hypothesis, for almost every $y$ and every $t > 0$, the original superlevel set is contained, up to a null set, in $J_y$. Hence $\mu(y,t) \le \ell(y)$, and (4.2) vanishes when $2|s| \ge \ell(y)$.

For nonnegative $f, g \in L^1(\mathbb{R})$, layer cake and the maximal intersection of centered intervals give

$$\int \min\{f^\star, g^\star\} = \int_0^\infty \min\{|\{f > t\}|, |\{g > t\}|\}\,\mathrm{d}t \ge \int \min\{f, g\}.$$

Together with mass preservation and $\|f-g\|_1 = \int f + \int g - 2\int \min\{f,g\}$, this gives $\|f^\star - g^\star\|_1 \le \|f-g\|_1$. Apply it to the fibers at $y+hu$ and $y$, and integrate in $y$. This proves (4.3); the difference-quotient limit in Lemma 3.1 proves the horizontal variation bound. □

The next identity computes the height retained by this rearrangement.

**Lemma 4.3** (Rearranged height)**.** *Let $\rho \in BV(\mathbb{R}^d)$ be a probability density and $v \in \mathbb{R}^d$. Set*

$$R(y,s) = \frac{1}{6}\rho(y+sv)\mathbf{1}_{\{|s|<3\}}, \tag{4.4}$$

*and rearrange $R(y,\cdot)$ symmetrically and decreasingly for almost every $y$, obtaining $R^\star$. Then*

$$\begin{aligned} \int |s| R^\star &= \frac{3}{2} - \frac{1}{24}\int_0^6 (6-h)\|\rho - \rho(\cdot + hv)\|_1\,\mathrm{d}h \\ &\ge \frac{3}{2} - \frac{3}{2}V_v(\rho). \end{aligned} \tag{4.5}$$

*Proof.* For a level $t > 0$, let $\mu(y,t) = |\{s \in (-3,3) : \rho(y+sv) > 6t\}|$, as in (4.2). The corresponding superlevel interval of $R^\star(y,\cdot)$ has first absolute moment $\mu(y,t)^2/4$. Layer cake and Tonelli give

$$\begin{aligned} \int |s| R^\star(y,s)\,\mathrm{d}y\,\mathrm{d}s &= \frac{1}{4}\int_{\mathbb{R}^d}\int_0^\infty \mu(y,t)^2\,\mathrm{d}t\,\mathrm{d}y \\ &= \frac{1}{24}\int_{-3}^3\int_{-3}^3\int_{\mathbb{R}^d} \min\{\rho(y+sv), \rho(y+rv)\}\,\mathrm{d}y\,\mathrm{d}r\,\mathrm{d}s. \end{aligned}$$

The change of level variable $\tau = 6t$ contributes the factor $1/6$. For $h \in \mathbb{R}$, the overlap

$$O(h) = \int \min\{\rho(y), \rho(y+hv)\}\,\mathrm{d}y = 1 - \frac{1}{2}\|\rho - \rho(\cdot+hv)\|_1 \tag{4.6}$$

is even in $h$. Integration over the square $(-3,3)^2$ therefore turns the preceding expression into $\frac{1}{12}\int_0^6 (6-h)O(h)\,\mathrm{d}h$, proving the identity. Only the probability-density hypothesis was needed for the exact identity. By (3.1), the density difference is at most $hV_v(\rho)$. Since $\int_0^6 h(6-h)\,\mathrm{d}h = 36$, this BV estimate gives the stated inequality. □

*Proof of Proposition 2.2.* Use the lift in Lemma 4.1 and the density $R$ in (4.4). Its mass is one and it vanishes almost everywhere outside $B$, with $V_{(u,0)}(R) = V_u(\rho)$ by translation invariance and (3.1). Lemma 4.2 gives a jointly measurable probability density $R^\star$, zero almost everywhere outside $B^\star$, with

$$V_{(u,0)}(R^\star) \le V_u(\rho) \le \kappa\|u\|_2. \tag{4.7}$$

The initial uniform marginal on $(-3,3)$ has mean absolute height $3/2$. Lemma 4.3 and (2.6) show that rearrangement loses at most $1/2$:

$$\int |s| R^\star \ge \frac{3}{2} - \frac{3}{2}\kappa\|v\|_2 \ge 1.$$

Apply Lemma 3.4 to $B^\star, R^\star$ at height one, using (4.7). By (4.1), this section is $\mathcal{T}_v K$. It is nonempty and supports a probability density with the same directional variation bound $\kappa$. □

*Remark* 4.4 (Normalization and the truncation height). For $v = 0$, the identity gives $\int |s| R^\star = 3/2$. For a one-dimensional equality example, take $v = 1$ and $\rho = (2A)^{-1}\mathbf{1}_{(-A,A)}$, where $A \ge 3$. Then $V_1(\rho) = 1/A$ and $\|\rho - \rho(\cdot + h)\|_1 = h/A$ for $0 \le h \le 6$. The mean absolute height is $3/2 - 3/(2A)$; at $A = 3$, it equals one and $V_1(\rho) = 1/3$.

The choice of cutoff three also has a geometric constraint. Let $K$ be nonempty, bounded, open, and convex, and take any $\rho \in \mathcal{P}(K)$. For $L > 1$, put $B_L = \{(y,s) : |s| < L,\ y + sv \in K\}$, and replace (4.4) by $R_L(y,s) = (2L)^{-1}\rho(y + sv)\mathbf{1}_{\{|s|<L\}}$. The same calculation gives

$$\int |s| R_L^\star = \frac{L}{2} - \frac{1}{8L}\int_0^{2L} (2L - h)\|\rho - \rho(\cdot + hv)\|_1 \, \mathrm{d}h \ge \frac{L}{2} - \frac{L^2}{6} V_v(\rho).$$

The height-one section of the symmetrized lift is $((K - v) \cap (K + v)) + \{tv : |t| < L - 1\}$. Containment in $(K - v) \cup (K + v)$ for every $K, v$ requires $L \le 3$: for $K = (-A, A)$, $A > 1$, and $v = 1$, the section is $(-A - L + 2, A + L - 2)$, whereas the union is $(-A - 1, A + 1)$. The argument of Lemma 2.1 proves sufficiency. For $2 < L \le 3$, a sufficient condition for the mean absolute height to be at least one is $V_v(\rho) \le 3(L - 2)/L^2$. The right-hand side increases with $L$ and equals $1/3$ at $L = 3$. This explains the parameter within the present uniform truncation and linear translation estimate; it does not assert an optimal stability threshold for other constructions.

# A Eigenvalue Convexity for the Regularized Norms

We give the eigenvalue argument needed in Lemma 3.2, following the infimal-convolution method of Wang and Xia [WX11, Section 3]. The analytic inputs are the eigenfunction results of Mosconi, Riey, and Squassina [MRS24]. The differential comparison is first made away from critical gradients; a cutoff then gives the global weak inequality, including the critical set. Throughout this appendix, the norm and $p > 1$ are fixed.

**Proposition A.1.** *Let $F$ be a norm on $\mathbb{R}^d$, smooth away from zero, with $D^2(F^2)$ positive definite there, and let $1 < p < \infty$. For nonempty bounded open convex sets $K_0, K_1$ and $0 \le t \le 1$, the eigenvalues defined by* (3.6) *satisfy*

$$\lambda_{p,F}((1 - t)K_0 + tK_1) \le (1 - t)\lambda_{p,F}(K_0) + t\lambda_{p,F}(K_1).$$

*Proof.* The cases $t = 0, 1$ are immediate. In dimension one, $F(\xi) = c|\xi|$ and the domains are intervals; scaling makes their eigenvalues a positive constant times the interval length to the power $-p$, which is convex. Assume henceforth that $d \geq 2$ and $0 < t < 1$. We first take smooth convex input domains and write $K_t = (1-t)K_0 + tK_1$ and $\lambda_i = \lambda_{p,F}(K_i)$.

*Eigenfunction inputs.* Set $a(0) = 0$ and, for $\xi \neq 0$,

$$\begin{aligned} a(\xi) &= F(\xi)^{p-1}\nabla F(\xi), \\ A(\xi) &= Da(\xi) = \frac{1}{p}D^2(F^p)(\xi) \\ &= F(\xi)^{p-2}\big((p-1)\nabla F(\xi)\otimes\nabla F(\xi) + F(\xi)D^2F(\xi)\big). \end{aligned}$$

Comparison with $D^2(F^2/2)$ and compactness of the unit sphere give constants $c_0, c_1 > 0$ such that

$$c_0|\xi|^{p-2}|\zeta|^2 \leq \zeta^{\mathsf{T}}A(\xi)\zeta \leq c_1|\xi|^{p-2}|\zeta|^2 \qquad (\xi \neq 0). \tag{A.1}$$

Also $F^p \in C^1(\mathbb{R}^d)$, since its gradient is $O(|\xi|^{p-1})$ at zero.

The direct method gives a nonnegative, nonzero first eigenfunction $u_i \in W_0^{1,p}(K_i)$. It minimizes

$$w \longmapsto \frac{1}{p}\int_{K_i} F(\nabla w)^p - \frac{\lambda_i}{p}\int_{K_i} w^p \qquad (w \geq 0),$$

whose minimum is zero by the definition of $\lambda_i$. In Theorem 1.1 of [MRS24], take the kinetic integrand to be $F^p$, the reaction to be $f(s) = \lambda_i s^{p-1}$, and its primitive to be $P(s) = \lambda_i s^p/p$. Then $P^{1/p}$ is linear, $P/f = s/p$ is convex, and the kinetic integrand is strictly convex. The transformation in that theorem is a positive multiple of $\log s$. Thus $u_i$ is log-concave; this specialization is also stated explicitly on p. 3674 of [MRS24]. The bounds (A.1) are condition (33) of that paper. With the extension $f(s) = \lambda_i|s|^{p-1}$, Proposition 4.5 of [MRS24] gives positivity and $C^{1,\alpha}$ regularity up to the smooth boundary for this nontrivial minimizer. In particular, $u_i$ has bounded gradient and zero boundary value. Where $\nabla u_i \neq 0$, the equation is locally uniformly elliptic with smooth coefficients, so local elliptic regularity gives $u_i \in C^2$.

Consequently $v_i = -\log u_i$ is convex and $C^1$ in $K_i$, is bounded below, and tends to infinity at the boundary. Away from $\{\nabla v_i = 0\}$, it is $C^2$ and satisfies

$$\operatorname{div} a(\nabla v_i) = \lambda_i + (p-1)F(\nabla v_i)^p. \tag{A.2}$$

This identity follows by differentiating $u_i = e^{-v_i}$ in the weak eigenfunction equation, which is classical on this region.

*Infimal convolution and its gradients.* For $z \in K_t$ and $\eta \geq 0$, define

$$w_\eta(z) = \inf_{\substack{x_i \in K_i \\ (1-t)x_0 + tx_1 = z}} \big((1-t)(v_0(x_0) + \eta|x_0|^2) + t(v_1(x_1) + \eta|x_1|^2)\big),$$

and put $w = w_0$. The infima are attained at interior pairs, because the $v_i$ are bounded below and blow up on their boundaries. When $\eta > 0$, the pair is unique by strict convexity. Since the input domains are bounded, $0 \leq w_\eta - w \leq C\eta$. For every compact $Z \subset K_t$, all minimizing pairs with $z \in Z$ and $0 \leq \eta \leq 1$ lie in fixed compact subsets of the input domains: an interior feasible pair remains feasible near each $z$ after shifting one component, and a finite cover bounds the objectives above. The lower bounds on the other component then bound each $v_i(x_i)$ above.

Stationarity gives a common gradient $q_\eta = \nabla v_i(x_i) + 2\eta x_i$. Convexity gives the supporting lower bound for $w_\eta(z + h)$, while shifting $x_0$ by $h/(1-t)$ gives the matching first-order upper

bound. Hence $w_\eta$ is differentiable and $\nabla w_\eta(z) = q_\eta$, also at $\eta = 0$, even if the minimizing pair is not unique. The compactness of minimizing pairs and continuity of $\nabla v_i$ now show that $w_\eta \in C^1(K_t)$ and

$$\nabla w_\eta \longrightarrow \nabla w \quad \text{uniformly on compact subsets of } K_t.$$

Indeed, a convergent sequence of output points and perturbation parameters has a subsequence of minimizing pairs converging to a minimizer for the limiting problem; its common gradient is uniquely determined by the output point.

As $z$ approaches $\partial K_t$, every limiting decomposition has a component on its input boundary. Thus $w(z) \to \infty$. The critical set $C = \{\nabla w = 0\}$ is therefore a nonempty compact convex subset of $K_t$. For $Z \Subset K_t \setminus C$, the common gradients stay bounded away from zero. Since $\nabla v_i(x_i) = q_\eta - 2\eta x_i$, the original gradients also stay away from zero for small $\eta$. The corresponding Hessians $D^2 v_i(x_i)$ are then uniformly bounded on these compact sets.

*The differential inequality away from the critical set.* At a perturbed minimizing pair, put $B_i = D^2 v_i(x_i) + 2\eta I$. On the compact sets just described, these matrices are positive definite, and implicit differentiation gives

$$D^2 w_\eta = \big((1-t)B_0^{-1} + tB_1^{-1}\big)^{-1} \le (1-t)B_0 + tB_1.$$

The inequality is the arithmetic–harmonic inequality for positive matrices. Taking its trace against $A(q_\eta) > 0$ and using (A.2) gives, with $\Lambda = (1-t)\lambda_0 + t\lambda_1$,

$$\operatorname{div} a(\nabla w_\eta) \le \Lambda + (p-1)F(\nabla w_\eta)^p + r_\eta, \qquad \sup_Z |r_\eta| \longrightarrow 0.$$

To justify the remainder, each original gradient differs from $q_\eta$ by $O(\eta)$, the Hessians are bounded, and $A$ is uniformly continuous on the compact annulus containing these gradients. The added Hessian contributes $2\eta \operatorname{tr} A$. The matrix $A$ has degree $p-2$; only its bounds on this annulus are used, also when $1 < p < 2$.

Put $U = e^{-w}$ and $U_\eta = e^{-w_\eta}$. Evenness and homogeneity give, at noncritical points,

$$\operatorname{div} a(\nabla e^{-v}) = e^{-(p-1)v}\big((p-1)F(\nabla v)^p - \operatorname{div} a(\nabla v)\big).$$

Thus $\operatorname{div} a(\nabla U_\eta) \ge -(\Lambda + r_\eta)U_\eta^{p-1}$ locally off $C$. Testing and passing to the limit using compact-uniform convergence of the functions and their first derivatives yields

$$\int_{K_t} a(\nabla U)\cdot\nabla\psi \le \Lambda \int_{K_t} U^{p-1}\psi \qquad (0 \le \psi \in C_c^\infty(K_t \setminus C)). \tag{A.3}$$

No convergence of second derivatives is needed.

*Including the critical set and the boundary.* The flux $a(\nabla U)$ is continuous and zero on $C$. Choose a Lipschitz cutoff $\chi_\delta$ which is zero within distance $\delta$ of $C$, one beyond distance $2\delta$, and has gradient bounded by $1/\delta$. Convexity of $C$ gives $\int |\nabla\chi_\delta| \le C_2$, uniformly for small $\delta$: the $2\delta$-parallel body has volume increment $O(\delta)$. This holds whether or not $C$ has interior. For $0 \le \phi \in C_c^\infty(K_t)$, insert $\psi = \phi\chi_\delta$ in (A.3), using smooth approximation away from $C$. The extra term has magnitude at most

$$C_2\|\phi\|_\infty \sup_{\operatorname{dist}(z,C)\le 2\delta} |a(\nabla U(z))| \longrightarrow 0.$$

Dominated convergence therefore gives

$$\int_{K_t} a(\nabla U)\cdot\nabla\phi \le \Lambda \int_{K_t\setminus C} U^{p-1}\phi \le \Lambda \int_{K_t} U^{p-1}\phi. \tag{A.4}$$

The first integral includes $C$ because its flux is zero. A positive-volume critical set causes no loss, since the right-hand integrand is nonnegative.

At any minimizing pair the common logarithmic gradient gives

$$U(z) = u_0(x_0)^{1-t} u_1(x_1)^t, \qquad |\nabla U(z)| = |\nabla u_0(x_0)|^{1-t} |\nabla u_1(x_1)|^t.$$

The bounded gradients of $u_i$ therefore make $U$ Lipschitz on the convex set $K_t$. If output points tend to its boundary, a subsequence of minimizing pairs has at least one component on its input boundary, so the product formula gives $U \to 0$. Consequently $(U - s)_+$ has compact support in $K_t$ for every $s > 0$, and these truncations converge to $U$ in $W^{1,p}$ as $s \downarrow 0$. Mollification proves $U \in W_0^{1,p}(K_t)$ and provides nonnegative smooth approximants. Testing (A.4) with $U$ now gives

$$\int_{K_t} F(\nabla U)^p \le \Lambda \int_{K_t} U^p.$$

Since $U > 0$, its Rayleigh quotient proves the desired inequality for smooth input domains.

*Arbitrary convex domains.* Every bounded open convex domain has nested smooth strongly convex inner exhaustions. For completeness, translate an interior point to zero and let $\gamma$ be its Minkowski functional. Convolution with a centered nonnegative smooth kernel gives a smooth convex function at least $\gamma$, converging uniformly on bounded sets. Adding $\delta|x|^2$ makes it strongly convex. Suitable regular sublevels below one lie compactly inside the domain and contain any prescribed compact subset. Successively including the preceding closure gives the required nested exhaustion.

Apply this to both inputs, obtaining $K_{i,j} \uparrow K_i$. Their Minkowski combinations increase to $K_t$. For every open exhaustion $L_j \uparrow L$, domain monotonicity and compactly supported smooth Rayleigh tests give $\lambda_{p,F}(L_j) \downarrow \lambda_{p,F}(L)$. Passing to this limit proves the assertion. □

All regularity and perturbation constants in this proof may depend on the fixed norm, $p$, the dimension, and the auxiliary domains. Lemma 3.2 subsequently takes $p \downarrow 1$ and $\epsilon \downarrow 0$ using scalar infima and fixed test densities; it requires no uniform eigenfunction estimates in either limit.

# References


[AS16] N. Alon and J. H. Spencer. *The Probabilistic Method*. Wiley, 4th edition, 2016. Section 13.2.

[AT26] D. J. Altschuler and K. Tikhomirov. Online Beck–Fiala down to logarithmic sparsity. arXiv:2607.14238v1, 2026.

[Ban98] W. Banaszczyk. Balancing vectors and Gaussian measures of $n$-dimensional convex bodies. *Random Structures & Algorithms*, 12(4):351–360, 1998. doi: 10.1002/(SICI)1098-2418(199807)12:4<351::AID-RSA3>3.0.CO;2-S.

[BDG19] N. Bansal, D. Dadush, and S. Garg. An algorithm for Komlós conjecture matching Banaszczyk's bound. *SIAM Journal on Computing*, 48(2):534–553, 2019. doi: 10.1137/17M1126795.

[BDGL19] N. Bansal, D. Dadush, S. Garg, and S. Lovett. The Gram–Schmidt walk: A cure for the Banaszczyk blues. *Theory of Computing*, 15(21):1–27, 2019. doi: 10.4086/toc.2019.v015a021.

[BF81] J. Beck and T. Fiala. "Integer-making" theorems. *Discrete Applied Mathematics*, 3(1):1–8, 1981. doi: 10.1016/0166-218X(81)90022-6.

[BJ26a] N. Bansal and H. Jiang. Decoupling via affine spectral-independence: Beck–Fiala and Komlós bounds beyond Banaszczyk. In *Proceedings of the 58th Annual ACM Symposium on Theory of Computing*, pages 432–442. ACM, 2026. doi: 10.1145/3798129.3800762. Cited version (full paper): arXiv:2508.03961v2.

[BJ26b] N. Bansal and H. Jiang. An exposition of the $\widetilde{O}(\log^{1/4} n)$ bound for the Komlós problem. arXiv:2608.28452v1, 2026.

[DGLN19] D. Dadush, S. Garg, S. Lovett, and A. Nikolov. Towards a constructive version of Banaszczyk's vector balancing theorem. *Theory of Computing*, 15(15):1–58, 2019. doi: 10.4086/toc.2019.v015a015.

[GK26] N. Guillen and V. A. Kobzar. Online Komlós converges to mean curvature flow. arXiv:2607.08943v1, 2026.

[KN08] B. Kawohl and M. Novaga. The $p$-Laplace eigenvalue problem as $p \to 1$ and Cheeger sets in a Finsler metric. *Journal of Convex Analysis*, 15(3):623–634, 2008.

[Kun23] D. Kunisky. The discrepancy of unsatisfiable matrices and a lower bound for the Komlós conjecture constant. *SIAM Journal on Discrete Mathematics*, 37(2):586–603, 2023. doi: 10.1137/22M149819X. Cited version: arXiv:2111.02974v1.

[LM15] S. Lovett and R. Meka. Constructive discrepancy minimization by walking on the edges. *SIAM Journal on Computing*, 44(5):1573–1582, 2015. doi: 10.1137/130929400. Cited version: arXiv:1203.5747v2.

[MRS24] S. Mosconi, G. Riey, and M. Squassina. Concave solutions to Finsler $p$-Laplace type equations. *Discrete and Continuous Dynamical Systems*, 44(12):3669–3697, 2024. doi: 10.3934/dcds.2024073.

[Nik13] A. Nikolov. The Komlós conjecture holds for vector colorings. arXiv:1301.4039v2, 2013.

[RR23] V. Reis and T. Rothvoss. Vector balancing in Lebesgue spaces. *Random Structures & Algorithms*, 62(3):667–688, 2023. doi: 10.1002/rsa.21113. Cited version: arXiv:2007.05634v2.

[Spe85] J. Spencer. Six standard deviations suffice. *Transactions of the American Mathematical Society*, 289(2):679–706, 1985. doi: 10.1090/S0002-9947-1985-0784009-0.

[SV26] G. Smirnov and R. Vershynin. Discrepancy and Fisher information. arXiv:2605.13107v1, 2026.

[Szu25] M. Szusterman. Monotonicity of the Gaussian measure under Banaszczyk transforms. arXiv:2510.00842v2, 2025.

[WX11] G. Wang and C. Xia. A Brunn–Minkowski inequality for a Finsler–Laplacian. *Analysis*, 31(2):103–115, 2011. doi: 10.1524/anly.2011.1073. Cited version: author manuscript.